BeeQ Report of Changes

Manuscript: BeeQ: Evaluating Quantum Fidelity Kernels as a New Approach Methodology for Molecular Toxicity Prediction

We thank the reviewers for their comments. This report details the implemented changes, their locations, and the technical scope of recommendations not fully implemented. The study focuses on comparing fixed quantum and classical kernels under matched conditions.

Page references refer to BeeQ_FinalV01.pdf, compared with BeeQ_Subm.pdf. Related weaknesses and recommendations are grouped. The responses follow Reviewer 1's five recommendations and Reviewer 2's two recommendations plus the third weakness.

Reviewer 1

Comment 1 Emphasis on toxic compound sensitivity

Reviewer comment. *Give the sensitivity drop on new data equal billing with the quantum-vs-classical comparison in the abstract and conclusion, it may be the more useful takeaway for readers who want to actually use this.*

Changes and response. We revised the closing synthesis of the Abstract and the opening of the Conclusions to state two main findings: the evaluated quantum kernels did not provide a consistent improvement over matched RBF, and sensitivity for toxic compounds decreased outside Development. The Introduction now identifies toxic-compound detection as an explicit purpose of the transfer assessment. Previously, the sensitivity decrease was reported, but the final synthesis emphasized kernel geometry.

Section III-B now adds the pooled out-of-fold sensitivity on Development, 0.338-0.410, alongside the historical-holdout range of 0.082-0.265, and reports that the models missed 36-45 of the 49 toxic compounds. These additions give the practical detection limitation greater prominence.

Location in the manuscript. Abstract, p. 1; Section I, final paragraph, p. 2; Section III-B, p. 5; Section IV, opening paragraph, p. 6.

Comment 2 Connection to kernel concentration

Reviewer comment. *Add a sentence connecting the results to the known "kernels flatten out" theory directly, rather than leaving readers to make that connection themselves.*

Changes and response. We retained the existing citation to Thanasilp et al. [4] in the Introduction and the geometric analysis in Section III-D, including the small-angle Gaussian approximation and the bandwidth-matched effective-rank comparison. The requested explicit discussion sentence was not added to the manuscript.

Location in the manuscript. Existing discussion: Section I, p. 1; Section III-D, pp. 5-6; reference [4], p. 8. No new concentration-specific paragraph was added.

Unimplemented elements and technical explanation. Because of the available paper space and the focus on a controlled comparison of fixed quantum and classical kernels, we deferred an expanded concentration analysis to future work. Technically, our experiments use one qubit count and exact statevector overlaps. High kernel alignment and similar effective ranks do not demonstrate exponential concentration as qubit count increases or the measurement-resource limitations studied in [4]. We therefore restrict the interpretation to similar geometry and predictive behavior in the evaluated setting.

Comment 3 Previous inspection of the historical holdout

Reviewer comment. *Flag the historical holdout's earlier exposure as a real limitation in the same breath as praising the provenance of the Costa Rica panel, so readers know to trust the two holdouts differently.*

Changes and response. We retained the explicit disclosure in Section II-C that the historical holdout had been inspected during earlier exploratory stages. The Introduction also continues to identify this earlier inspection when presenting the transfer evaluations. These disclosures were already present in the previous version; an expanded account of the exploratory history was not added.

Location in the manuscript. Retained disclosure: Section I, final paragraph, p. 2, and Section II-C, p. 2. The final fitting and threshold-selection procedure is in Section II-D, p. 3.

Unimplemented elements and technical explanation. Paper-space constraints and the emphasis on the quantum-versus-classical comparison led us to defer the broader methodological account. This does not remove the limitation: cluster-disjoint partitions and Development-only final fitting cannot undo possible influence from earlier inspection on upstream decisions. The historical holdout must therefore be interpreted as a complementary transfer assessment rather than an untouched external confirmation. CR8 provides a separate external challenge, but its small size also limits generalization. The detailed chronology and extent of any earlier influence remain unresolved; we do not claim that such influence was absent.

Comment 4 Methods for adapting quantum kernels

Reviewer comment. *Mention existing methods for fixing or adjusting quantum kernels so they don't flatten out, since the conclusion already points in that direction without citing anything specific.*

Changes and response. We replaced the broad future-work paragraph with concrete directions: extending encoding-scale optimization, training embedding parameters through kernel-target alignment, and evaluating projected kernels based on local quantum observables. The revised text proposes initially keeping X10 fixed to distinguish kernel changes from representation changes and evaluating these approaches under nested molecular-cluster validation, with attention to toxic-compound recovery and false positives.

We added Shaydulin and Wild, "Importance of kernel bandwidth in quantum machine learning," as reference [28]. The present contribution remains the comparison of fixed kernels. Training or testing adaptive and projected kernels is reserved for future work because it would introduce additional model-selection choices beyond that controlled comparison; no performance improvement from these methods is claimed.

Location in the manuscript. Section IV, final paragraph, p. 7; new reference [28], p. 8.

Comment 5 MLP class balancing and interpretation

Reviewer comment. *Rerun the MLP model with the same class balancing used for the others or clearly flag the mismatch as a limitation before attributing its worse results to the model itself.*

Changes and response. We performed a retrospective MLP balancing analysis using X10 and the frozen molecular-cluster partitions. Inverse-frequency sample weights were computed within each training fit and passed to the MLP; scaling and evaluation remained unweighted. The analysis includes fixed-hyperparameter and fully retuned variants. The revised manuscript incorporates the latter, using the same 12-candidate, five-outer/four-inner protocol.

In Table I, the MLP AUROC changes from 0.677 ± 0.125 to 0.679 ± 0.100, AUPRC from 0.596 ± 0.138 to 0.594 ± 0.112, balanced accuracy from 0.654 to 0.632, and MCC from 0.428 to 0.338. Section II-D adds

training and search details. Section III-B adds holdout AUPRC 0.329, detection of 13 of 49 toxic compounds, and 22 false positives among 132 non-toxic compounds. The project records confirm that the original MLP detected 4 of 49 toxic compounds with 4 false positives. Increased toxic recovery therefore came with a substantial specificity cost; balancing did not resolve weak discrimination.

Location in the manuscript. Section II-D, p. 3; Table I, p. 4; Section III-B, p. 5; Section III-E, p. 6. Supporting project records: MLP sensitivity report and full_20261009_v2 configuration and results.

Remaining reporting limitations. These analyses are retrospective because the external results were already known. We do not attribute the original MLP performance solely to its architecture or present the new evaluation as blind validation. Manuscript integration remains incomplete: II-D does not yet state the weighting explicitly, and III-B retains some original MLP values alongside the updated results. Those descriptions require reconciliation.

Reviewer 2

Comment 1 Quantitative context for X10

Reviewer comment. *Benchmark X10 against prior ApisTox results. Include a direct quantitative comparison (even a summary table) of BeeQ's AUROC/AUPRC against previously published fingerprint-, graph kernel-, and GNN-based models on ApisTox, so readers can judge whether the compact 10-descriptor representation is competitive or represents a deliberate trade-off in interpretability/qubit-compatibility versus raw performance.*

Changes and response. We added Section III-F and Table II to provide quantitative context for X10. The table reports AUROC for WL-OA with SVM (0.805 ± 0.073), ECFP with Random Forest (0.787 ± 0.091), RDKit2D with Random Forest (0.784 ± 0.064), Mordred with Random Forest (0.755 ± 0.103), atom counts with Random Forest (0.748 ± 0.068), and MACCS with Random Forest (0.732 ± 0.104), alongside X10 with Random Forest (0.748 ± 0.057). The earlier manuscript cited the ApisTox benchmark without this quantitative comparison.

The added text explains the deliberate compactness of X10 and its one-descriptor-per-qubit design. This comparison places its predictive performance in context without claiming that X10 is the strongest molecular representation.

Location in the manuscript. New Section III-F, p. 6; new Table II, p. 7; retained reference [9], p. 8.

Unimplemented elements and technical explanation. The table reports AUROC only and contains no GNN row, so this recommendation is partially implemented. A direct ranking against published ApisTox results would require accounting for the curated cohort, partitions, preprocessing, and evaluation protocol; BeeQ893 contains 893 compounds derived from the 1,035-compound source dataset. We therefore do not interpret the table as evidence of superiority over published GNNs or as a fully harmonized external benchmark. The provenance of each added value and the omitted AUPRC/GNN comparisons still require clearer reporting.

Comment 2 Representations beyond X10

Reviewer comment. *Test representation dimensionality beyond X10. Run the same pipeline with an expanded descriptor set (e.g., X15, X20, or a full fingerprint baseline) to disentangle whether the observed sensitivity limitations and quantum-classical equivalence stem from the learning method or from the compact representation forced by the 10-qubit constraint.*

Changes and response. The new Section III-F and Table II extend the reported classical comparison beyond X10 to fingerprints, descriptor families, atom counts, and a graph kernel. In particular, ECFP, RDKit2D, and Mordred provide alternatives to the compact descriptor set. The accompanying discussion explains why the quantum comparison retains ten descriptors and ten qubits.

Location in the manuscript. New Section III-F, p. 6; new Table II, p. 7; Section IV, final paragraph, p. 7.

Unimplemented elements and technical explanation. We did not add X15/X20 quantum experiments or transfer and sensitivity analyses for the expanded representations. Keeping X10 fixed isolates the kernel comparison; in this encoding, adding descriptors also adds qubits. A broader dimensionality study was left for future work within the paper-space and scope limits. The added AUROC comparison does not establish whether X10 causes the sensitivity loss or the similar quantum and classical performance.

Comment 3 Scope of conclusions from CR8

Reviewer comment. *CR8 conclusions rest on n=2 toxic compounds. While the authors appropriately caveat this in the Results, the Conclusion section restates the CR8 failure with a rhetorical weight that outpaces what two data points can statistically support, even though it remains a valid qualitative observation.*

Changes and response. We strengthened the small-sample qualification in Section III-E and the Conclusions. Section III-E now explicitly limits CR8 to compound-level observations rather than population-level conclusions. The Conclusions state that eight compounds, including only two toxic compounds, do not support an estimate of population sensitivity. This extends the caution previously stated in the Results.

The updated Section III-E reports that the balanced MLP identified Isocycloseram and produced false positives for Fluoxapiprolin and Florylpicoxamid. These results match the retrospective analysis. The original unweighted MLP missed both toxic compounds and produced one false positive. Recovering one of two toxic compounds does not establish reliable population sensitivity.

Location in the manuscript. Section III-E, p. 6; Section IV, p. 7. The remaining consistency issues concern the Abstract, p. 1, and Figure 4 and Section IV, p. 7.

Remaining reporting limitations. The Abstract, Conclusions, and Figure 4 still retain the original all-model failure account without consistently distinguishing it from the balanced MLP results. The statistical qualification was implemented, but these descriptions and the figure require final reconciliation. We do not claim that this reporting inconsistency has already been corrected.

Additional editorial changes

The revision replaces anonymous author placeholders with names, affiliations, and email addresses and adds the conference heading. Equations (2)-(4) are numbered, and the unresolved cross-reference in Section III-D has been replaced by a reference to Equation (4).